SISV_1% polymer (THF/DOX 7:3) — provisional DLS analysis report

Executive Summary

Dataset. Polymer in THF/dioxane by multi-angle PCS across 7 temperatures (11.5–30.5 °C); each is a MAPCS experiment (27 angles × {VH,VV,3D}, 5 replicates). The full suite runs on VV at every temperature.

Main result. A narrow, ˜120 nm, nearly optically isotropic polymer. The z-average R_h appears to fall 139→106 nm over 11.5→30.5 °C, but this is mostly the **fixed-viscosity artifact** (η recorded as a constant 0.47 mPa·s); the same-sample solve implies the true η drops ˜24% so the size is essentially T-independent. PDI stays ˜0.26 and the predictive CV R^2 is ˜1.0 (clean monodisperse). $\rho = R_g/R_h$ **is not determined** (low-q structure factor, not a Guinier R_g); the static intensity shows a weak transition near 15–22 °C.

Confidence. The z-average r_h carries replicate/angle bootstrap intervals; leave-one-angle-out cv gives predictive r^2=1.000; the corenn inversion converged.

Assumption-sensitive. Viscosity, refractive index, q-from-angle calibration, the toluene/absolute-intensity reference, the assumed sphere form factor, and the CORENN regularization all enter the reported sizes; see the Assumption Register.

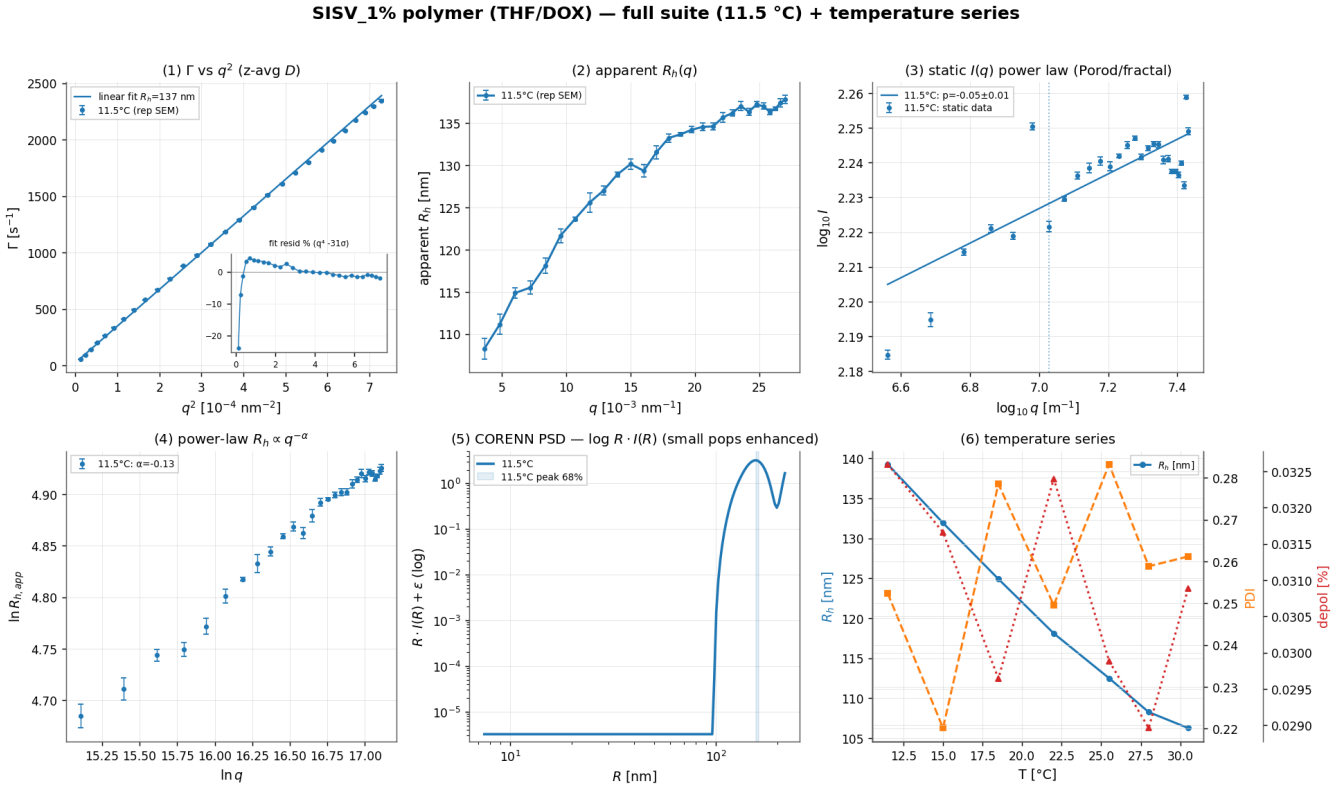
Recommended next action. supply the true η(T) of THF/dioxane to remove the apparent-shrinkage artifact; confirm the toluene reference to lock the SLS scale.

> **Provisional flags:** viscosity is a fixed placeholder at all T (apparent R_h(T) is mostly artifact); toluene/absolute-intensity reference not confirmable from the export; sphere form factor assumed (ρ not determined — low-q structure factor).

Summary Table

T [°C]	$R_{h,z}$ [68% CI nm]	CV R^2	PDI	α	CORENN peak	I_{VV}	depol [%]	η/η_lo
11.5	139 [139–140]	1.000	0.25	-0.13	157 nm	174	0.033	1.00
15.0	132 [132–132]	1.000	0.22	-0.11	145 nm	153	0.032	0.95
18.5	125 [124–125]	1.000	0.28	-0.18	127 nm	154	0.030	0.90
22.0	118 [118–118]	1.000	0.25	-0.15	130 nm	156	0.032	0.85
25.5	113 [112–113]	1.000	0.28	-0.16	103 nm	162	0.030	0.81
28.0	108 [108–109]	1.000	0.26	-0.13	106 nm	173	0.029	0.78
30.5	106 [106–107]	1.000	0.26	-0.13	107 nm	172	0.031	0.76

Consolidating Figures



Consolidating figure: full suite at 11.5 °C (5 panels) plus the temperature series of R_h/PDI/depolarization (raw-derived).

Primary Model Results

Detailed suite shown at the representative temperature (11.5 °C); all temperatures are tabulated below and in the per-temperature panels.

Step 1 — per-angle cumulants → Γ(q) (VV). A 2nd-order cumulant on the initial decay (g1_min=0.6) gives the decay rate at each angle; 28 angles are kept.

Step 2 — Γ vs q^2 (apparent diffusion). A straight-line fit $\Gamma = c + D \cdot q^2$ over all 28 angles (no high-q rolloff detected) gives the **apparent diffusion coefficient D = 3.247e-12 ± 2.4e-15 m^2/s → R_h = 137 nm** (r^2=1.000) [replicate/repeatability 68% 139–139; angle/model 68% 139–140 nm]. **The fit residuals are structured, not random** (figure inset): they trend systematically with q (Spearman resid–q = -0.70; runs 3 vs ˜15 expected) and a q^4 curvature term is significant at **-31σ** — so even with the rolloff removed, Γ vs q^2 genuinely **curves downward** — D_{app} falls with q, a non-diffusive low-q offset (slow large-aggregate mode, weak S(q), or baseline): the line is a chord and no single slope is *the* diffusion coefficient. The q-dependent apparent size (next step) and the CORENN PSD are the proper description.

Step 3 — apparent R_h(q) (polydispersity). apparent R_h rises with q, from **115 nm** (low q) to **137 nm** (high q) (curvature C=-2.43e-16, Spearman +0.99); mean PDI 0.25 — a narrow population with weak interparticle structure (S(q)).

Step 4 — static intensity I(q), regime from the PSD. The CORENN PSD gives dominant scatterers of R ≈ 157 nm (R_g ≈ 122 nm, sphere-derived from the PSD radius — no Guinier regime is reached, so this is a model estimate, not a measured R_g), so q·R_g spans 0.4–3.3 across the angles → **spans Guinier (low-q) and Porod (high-q)**. The 6 low-q points are in the Guinier window but the intensity does not fall (no positive R_g) — a low-q structure factor

S(q) < 1 / interparticle correlations rather than a particle Guinier radius. On the 28 point contiguous lower-q scaling window (q=0.00365–0.027 nm^{-1}): a near-flat / noisy high-q intensity ($I \propto q^{+0.0}$), so no clean Porod slope (r^2=0.691) (reported as diagnostic only).

Step 5 — dynamic q-dependence. The apparent size follows a power law $R_{h,app} \propto q^{\hat{}}\text{-}\alpha$ with α=-0.13 (r^2=0.99) — a compact descriptor of the breadth of the distribution ($\alpha \to 0$ monodisperse, $\alpha \to 1$ fully scale-free).

Step 6 — CORENN multi-angle PSD (forward model). The joint inversion of the resolvable angles, on a radius grid bounded by dynamic observability plus the static q-window (PSD grid 8–217 nm, upper bound set by dynamic) and the main-relaxation lag window (λ=1.1e+02), recovers one q-consistent intensity size distribution: peak(s) at **157 nm (100%)**. An 20-angle bootstrap of the inversion puts the primary peak at 158 nm [68% 156–162; 95% 152–164 nm].

Static Regime And Shape

The CORENN PSD anticipates which static regime each angle probes (Guinier where $q \cdot R_g < 1.3$, Porod where $q \cdot R \gtrsim \pi$). When a **valid Porod slope** exists, its exponent informs the form-factor / morphology model (q^{-4} a smooth sphere; 3<p<4 a surface fractal, D_s=6-p; p≈3 a rough-surface or branched network; 2≤p<3 a mass-fractal / branched aggregate, D=p; p≈2 a coil; p≈1 a rod). $\rho = R_g/R_h$ is the model-independent shape check (sphere ≈ 0.78, coil ≈ 1.5, rod $\gtrsim$ 2), determined only when a static Guinier R_g exists.

condition	$q \cdot R_g$ span	regime	static fit	implied morphology	$\rho = R_g/R_h$
11.5 °C	0.4–3.3	spans Guinier (low-q) and Porod (high-q)	flat/rising $I \propto q^{+0.0}$ (no clean slope)	—	not determined
15.0 °C	1.2–3.0	form-factor / Porod ($q \cdot R \gtrsim \pi$ throughout)	flat/rising $I \propto q^{+0.0}$ (no clean slope)	—	not determined
18.5 °C	0.4–2.6	spans Guinier (low-q) and Porod (high-q)	flat/rising $I \propto q^{-0.1}$ (no clean slope)	—	not determined
22.0 °C	0.5–2.7	spans Guinier (low-q) and Porod (high-q)	flat/rising $I \propto q^{-0.1}$ (no clean slope)	—	not determined
25.5 °C	0.3–2.2	spans Guinier (low-q) and Porod (high-q)	flat/rising $I \propto q^{+0.0}$ (no clean slope)	—	not determined
28.0 °C	0.3–2.2	spans Guinier (low-q) and Porod (high-q)	flat/rising $I \propto q^{-0.1}$ (no clean slope)	—	not determined
30.5 °C	0.3–2.2	spans Guinier (low-q) and Porod (high-q)	flat/rising $I \propto q^{+0.1}$ (no clean slope)	—	not determined

Particle model. The PSD assumes a homogeneous-sphere form factor. No valid Porod slope and ρ is **not independently determined** (no Guinier regime, or a low-q structure factor), so the sphere assumption is not confirmed; treat the PSD radii as model-dependent.

Error Propagation And Sensitivity

Headline R_h and the PSD peak carry bootstrap intervals (ρ where determined). The **replicate** bootstrap is measurement repeatability; the **angle/model** bootstrap is robustness to angle selection — the dominant term for polydisperse samples.

condition	R_h replicate 68%	R_h angle/model 68%	PSD peak 68%	ρ 68%
11.5 °C	139–139 nm	139–140 nm	156–162 nm	—
15.0 °C	132–132 nm	132–132 nm	—	—
18.5 °C	125–125 nm	124–125 nm	—	—
22.0 °C	118–118 nm	118–118 nm	—	—
25.5 °C	112–113 nm	112–113 nm	—	—
28.0 °C	108–108 nm	108–109 nm	—	—
30.5 °C	106–106 nm	106–107 nm	—	—

Systematic sensitivity (not included in the intervals above). $R_h = \mathrm{kT}/(6\pi\eta\mathrm{D})$ scales inversely with viscosity, so a viscosity error propagates linearly into R_h; the q calibration (scattering angle + refractive index) shifts every q-dependent quantity; the absolute-intensity (toluene) scale sets the SLS magnitude; and the assumed sphere form factor biases the PSD radii for non-spherical particles. See the Assumption Register.

Fit Quality And Model Adequacy

Beyond r^2: a hold-out cross-validation (predictive R^2), a residual-structure test (runs / lag-1 autocorrelation — structured residuals flag model inadequacy even at high r^2), and the CORENN inversion's convergence and conditioning.

condition	Γ-vs-q^2 R^2 (in / CV)	Γ resid	CORENN g2 resid	CORENN: conv / GCV / κ / eff-DOF
11.5 °C	1.000 / 1.000	**structured**	**structured**	yes / interior / κ=5e+11 / 13/24
15.0 °C	1.000 / 1.000	ok	**structured**	yes / interior / κ=9e+11 / 15/24
18.5 °C	1.000 / 1.000	**structured**	**structured**	yes / interior / κ=1e+10 / 12/24
22.0 °C	1.000 / 1.000	**structured**	**structured**	yes / interior / κ=4e+11 / 15/24
25.5 °C	1.000 / 1.000	**structured**	**structured**	yes / interior / κ=3e+09 / 11/24
28.0 °C	1.000 / 1.000	**structured**	**structured**	yes / boundary / κ=2e+09 / 10/24
30.5 °C	1.000 / 1.000	**structured**	**structured**	yes / boundary / κ=2e+10 / 11/24

Structured Γ-vs-q^2 residuals reflect the known polydispersity curvature (the single slope is a summary, not the model); structured CORENN g2 residuals flag form-factor / single-distribution imperfection. A predictive CV R^2 well below the in-sample R^2 indicates over-fitting / poor extrapolation across angles.

Conditioning: $\kappa \approx$ 9e+11 means the inversion is ill-conditioned (˜12 of ˜16 double-precision digits lost), so the recovered distribution is fixed by the regularization, not the data alone — reinforcing the modality caveat. The eff-DOF/n_{basis} above is the multi-angle inversion's data-supportable complexity; it is a different inversion from the single-angle stitched one whose eff-DOF (˜5–6) appears in the size_max diagnostic.

Sample Synthesis

What the independent observables imply about the sample as a colloid — derived only from the scattering data above, with the inversion caveats carried through:

Physical picture. The scattering is dominated by ˜**156 nm** objects (intensity-weighted); by number the population centres near ˜**142 nm**, and the z-average R_h is 139 nm — a single, fairly narrow population — intensity and number weightings nearly agree (see the size-distribution-by-weighting and size_max sections).

Hydrodynamic shape (Guinier–Porod → Stokes–Einstein). The static slope identifies the dominant scatterer

as **a sphere (assumed)**, so the geometry-matched Stokes–Einstein relation is $\mathbf{D = kT/(6\pi\eta R)}$ — the radius R = R_h is **137 nm**. No resolved Porod slope, so the sphere form is assumed.

Primary unit. The number-weighted mode (˜142 nm) is the smallest resolved scatterer — plausibly the primary particle or smallest aggregate.

Caveats that bound these conclusions. *(1) Dilute limit / S(q):* the analysis assumes single-particle scattering; if the suspension is not dilute, a low-q structure factor (interparticle correlations or incipient gelation) can bias the Porod/fractal slope and the apparent sizes — a dilution series would test it. *(2) Inversion dependence:* the modes and the aggregation number inherit the ill-posedness and size_max / λ / form-factor sensitivity documented in the size-distribution sections. *(3) Stability:* this is a single time-point of a system that can aggregate or sediment; a repeat / time-resolved run (or checking the count rate for drift across the angle scan) is needed to establish the distribution is stationary.

Consistency across observables. The narrow dynamic distribution (mean PDI 0.25) is consistent with the single population seen across all weightings. No single observable here is contradicted by another — the picture is one well-defined, fairly narrow population.

Mode-Specific Diagnostics

Temperature series (kinetics)

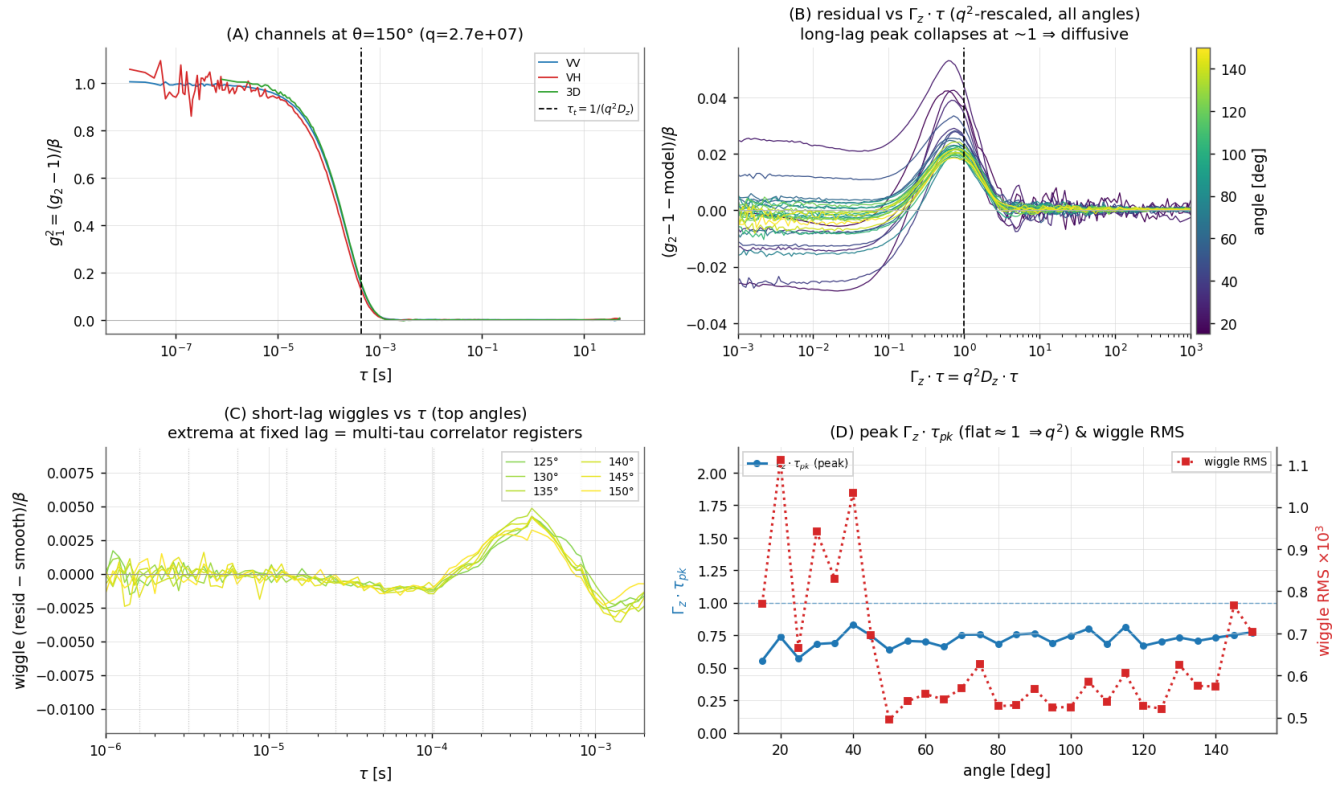
- **z-average** R_h falls smoothly 139→106 nm from 11.5→30.5 °C (but see the viscosity caveat below).
- **PDI** stays narrow and stable (mean 0.26, 0.22–0.28) — a well-defined single population.
- **static intensity** is non-monotonic: a minimum near 15.0 °C (153 kHz/mW) then a rise above ˜22 °C — a candidate conformational/aggregation transition.

Polarization (VV/VH depolarized)

The depolarization ratio $I_{\mathrm{VH}}/I_{\mathrm{VV}}$ is very small (mean 0.031 %), i.e. the polymer is nearly optically isotropic. Splitting translational from rotational diffusion (Γ_VV=q^2D_t, Γ_VH=q^2D_t+6D_r) gives a small, noisy VH intercept: for this ˜120 nm polymer rotational diffusion ($D_r \propto 1/\mathrm{R}^3$) sits near the resolution limit, so the depolarization ratio is the robust observable.

High-angle correlogram diagnostics (μs–ms ripples)

The high-angle single-exponential residual contains **two distinct features** that separate by their q-scaling: a **long-lag peak** that shifts to shorter lag with increasing angle, and **short-lag wiggles** that barely move. Panels (B)–(D) tell them apart across all angles.



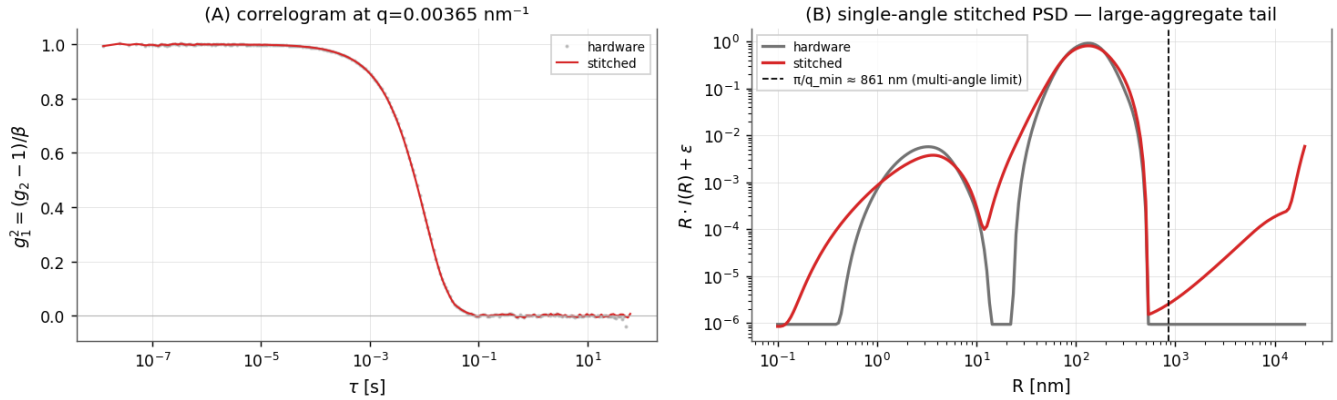
Reading the panels. **(A)** VV/VH/3D at the top angle (a faster VH would be the rotational signature Γ_VH=q$^2 D_t$+6D_r; a VV-vs-3D split flags modulation / multiple scattering). **(B)** residual vs the q^2-rescaled lag Γ_z·τ — a feature that **collapses onto Γ_z·τ≈1** for every angle is a genuine diffusive relaxation. **(C)** the short-lag wiggle component vs *absolute* τ — extrema that line up at **fixed lag** (dotted: the multi-tau correlator register boundaries) are instrumental, not q-dependent dynamics. **(D)** trends: a flat Γ_z·τ_pk ≈ 1 confirms the peak scales as q^2; the wiggle RMS grows as the signal weakens at high q.

This sample — the two features separated across all 28 angles: **(1) the long-lag peak** sits at Γ_z·τ_pk ≈ 0.74 and stays roughly constant across angles (panel D), i.e. it tracks τ_pk = 1/(q$^2 D_z$) ∝ 1/q^2 and collapses onto a single curve under q^2 rescaling (panel B) — a **genuine diffusive multi-mode / polydispersity** deviation (the broad relaxation-time distribution of the sample), strongest at high q. **(2) the short-lag wiggles** pin to *fixed* absolute lags (≈ 13, 26, 51, 102 μs), independent of angle and locked to the multi-tau correlator's register boundaries (panel C), with the amplitude growing ≈1× from low to high angle as the signal weakens — a **correlator / instrumental artifact**, not sample dynamics.

Channel check (A): at the top angle Γ_VH/Γ_VV = 1.17, i.e. VH decays faster (a rotational signature *if* not VH shot noise); but the depolarized channel is weak ($I_{\mathrm{VH}}/I_{\mathrm{VV}}$ ≈ 0.0 %) so Γ_VH is noise-limited.

Slow-mode / large-aggregate tail (single-angle stitched)

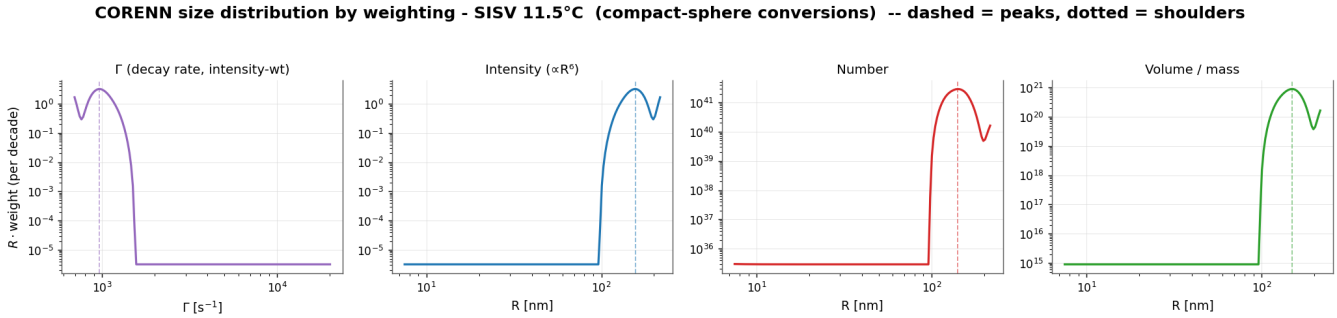
The multi-angle intensity PSD is q-resolution-limited: its R^6-weighted q→0 representation, coupled through the sphere form factor, edge-piles for R $\gtrsim \pi/q_{\min}$ ≈ **861 nm**. A *single-angle* CORENN inversion of the gap-aware **stitched** correlogram (clean long-lag pooled over the full run, τ_max 50→61 s) instead recovers the slow diffusive modes directly from the decay-rate spectrum — with a data-bounded amplitude, so it resolves the large-aggregate tail where the multi-angle PSD cannot.



At the lowest angle (q=0.00365 nm^{-1}), the stitched single-angle inversion finds **negligible intensity (<2%) beyond** $\pi/q_{\min}$ (extent ˜273 nm) — there is no resolvable large-aggregate tail; the sample is a single well-defined population, consistent with the multi-angle PSD.

Size distribution by weighting — continuous summary, then apparent modes

DLS recovers a size distribution by an **ill-posed Laplace inversion**: many different distributions — a smooth continuum or a set of discrete modes — reproduce the measured correlation function equally well within noise, and the regularized inversion (CORENN) returns just one of them, fixed by its smoothness penalty, radius grid, and form-factor assumption. The number and sharpness of peaks are therefore properties of the inversion, not facts the data establish on their own. The report gives **two views**: (1) assumption-light continuous percentiles, and (2) the inversion's apparent modes/shoulders, flagged as inversion-dependent and trustworthy only where they survive the size_max / λ sensitivity tests. CORENN returns the *same* inversion under every weighting (the vendor's Γ / Intensity / Number / Volume tabs); re-weighting only relocates and re-weights the features.



Γ (decay-rate) view — the rawest signal (figure panel 1). Before any size model, the dominant intensity component relaxes at $\Gamma \approx$ **964 s^{-1}** (relaxation time $\tau = 1/\Gamma \approx$ **1.04 ms**) at the reference angle; the size columns below are this spectrum mapped through Stokes–Einstein $\Gamma = k_B T\ q^2/(6\pi\eta R)$.

Continuous-distribution summary (percentiles; no modes assumed). The assumption-light description — percentiles of the recovered distribution, robust to how it is partitioned into modes. Intensity is narrow (dominated by the largest scatterers); number-weighting (÷R^6) strongly amplifies the inverted PSD's small-R end, so its percentiles are the least certain.

Weighting	D10 (nm)	D50 (nm)	D90 (nm)	D_g (nm)	σ_g (GSD)	span	log-skew
Intensity (∝R^6)	131	155	185	156	1.15	0.35	+0.25
Number (sphere, R^6)	119	141	165	141	1.13	0.33	+0.19
Volume (sphere, R^3)	125	148	173	148	1.14	0.33	+0.21

Modality caveat — read the next table as inversion fine structure. The Laplace inversion is ill-posed: a smooth continuum and a set of discrete modes can fit the same g_2 within noise, so the number and sharpness of peaks depend on the regularization (λ), the radius grid / size_max, the spline basis, and the sphere form factor — none of which the data uniquely fix. The inversion carries ≈ **13 spline degrees of freedom** — ample latitude to place peaks on a continuum. **Treat the apparent peaks/shoulders below as fine structure of this particular inversion — not established sub-populations — unless they are stable under the size_max / λ sensitivity checks and reproducible across replicates.**

Apparent modes & shoulders (CORENN fine structure — inversion-dependent)

Weighting	Comp.	Peak R (nm)	⟨R⟩ (nm)	Median (nm)	Fraction	GSD	log-skew
Intensity ($\propto R^6$)	1	156	157	155	100%	1.15	+0.25 (≈ log-symmetric)
Number (sphere, R^6)	1	142	142	141	100%	1.13	+0.19 (≈ log-symmetric)
Volume (sphere, R^3)	1	150	149	148	100%	1.14	+0.21 (≈ log-symmetric)

The intensity distribution is led by a peak at **156 nm** (100% of scattered light, ≈ log-symmetric); under number weighting the dominant component is at **142 nm** — the *same* data, modes relocated and re-weighted by the R-power, which is why no single mean describes it.

Volume/Number assume compact spheres ($V \propto R^3$); no mass-fractal correction is applied (the static slope is consistent with a compact / surface-fractal object, $D_f \approx 3$).

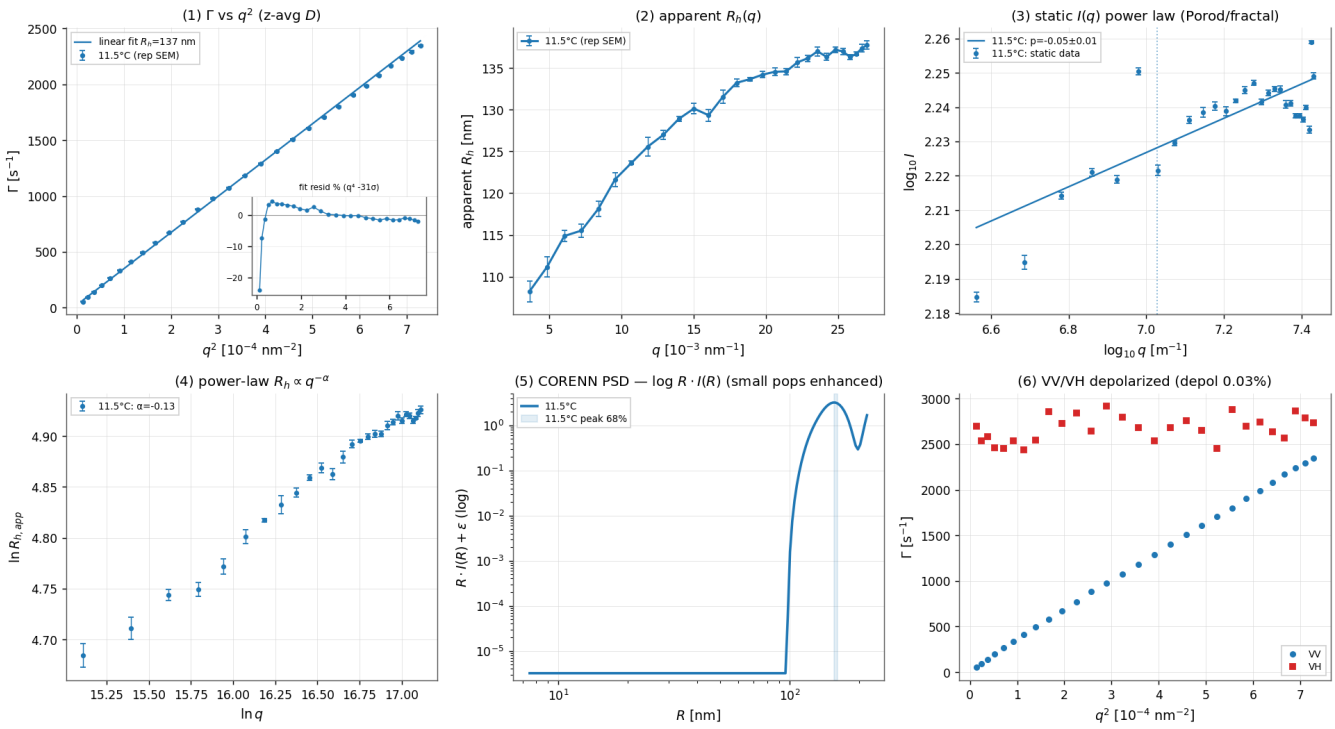
Joint multi-temperature (same sample)

The recorded viscosity is a fixed 0.47 mPa·s at every temperature (THF actually thins on heating). Assuming a T-invariant polymer size, the implied viscosity drops **24 %** over 12→30 °C — consistent with THF — so the apparent R_h decline is mostly the fixed-viscosity artifact, not real shrinkage.

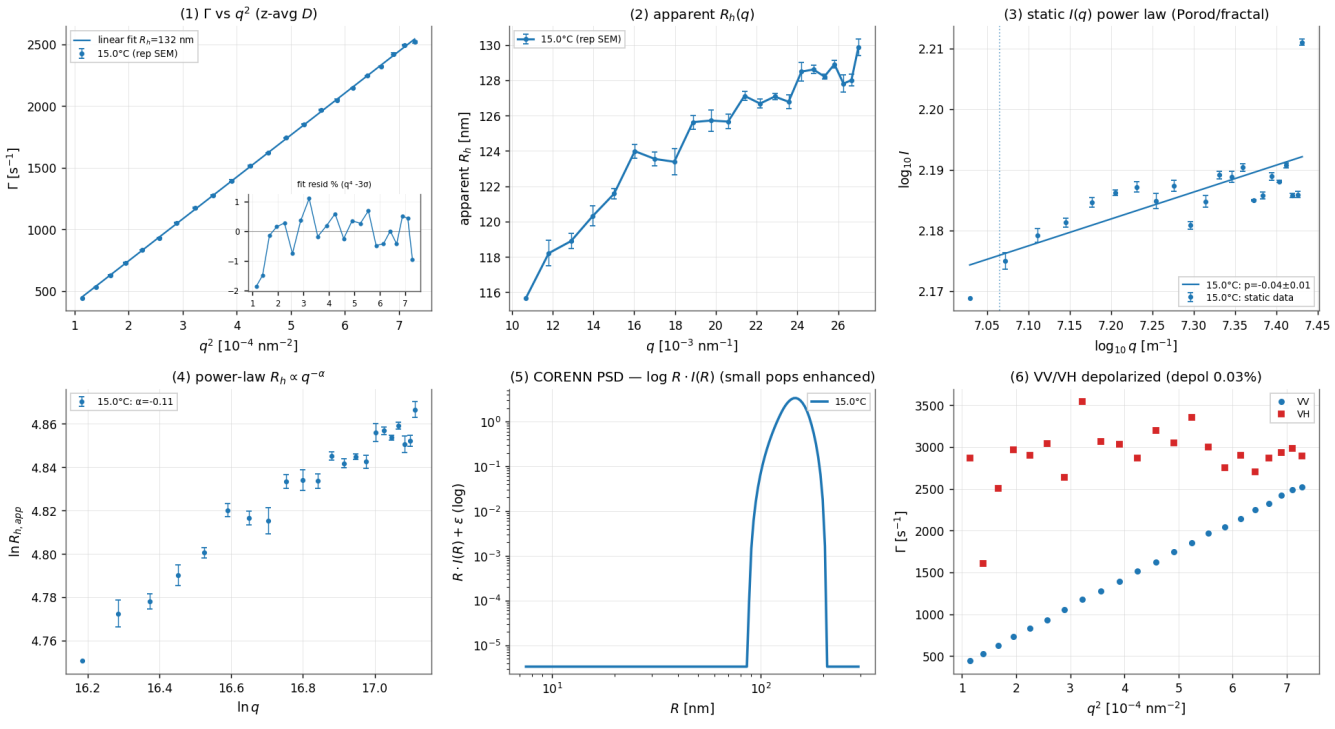
All analysis panels, at every temperature

The complete static+dynamic suite (Γ-vs-q^2, apparent $R_h(q)$, regime-aware static I(q), $R_h \propto q^{\hat{}}\text{-}\alpha$, CORENN PSD, VV/VH depolarized) at each temperature:

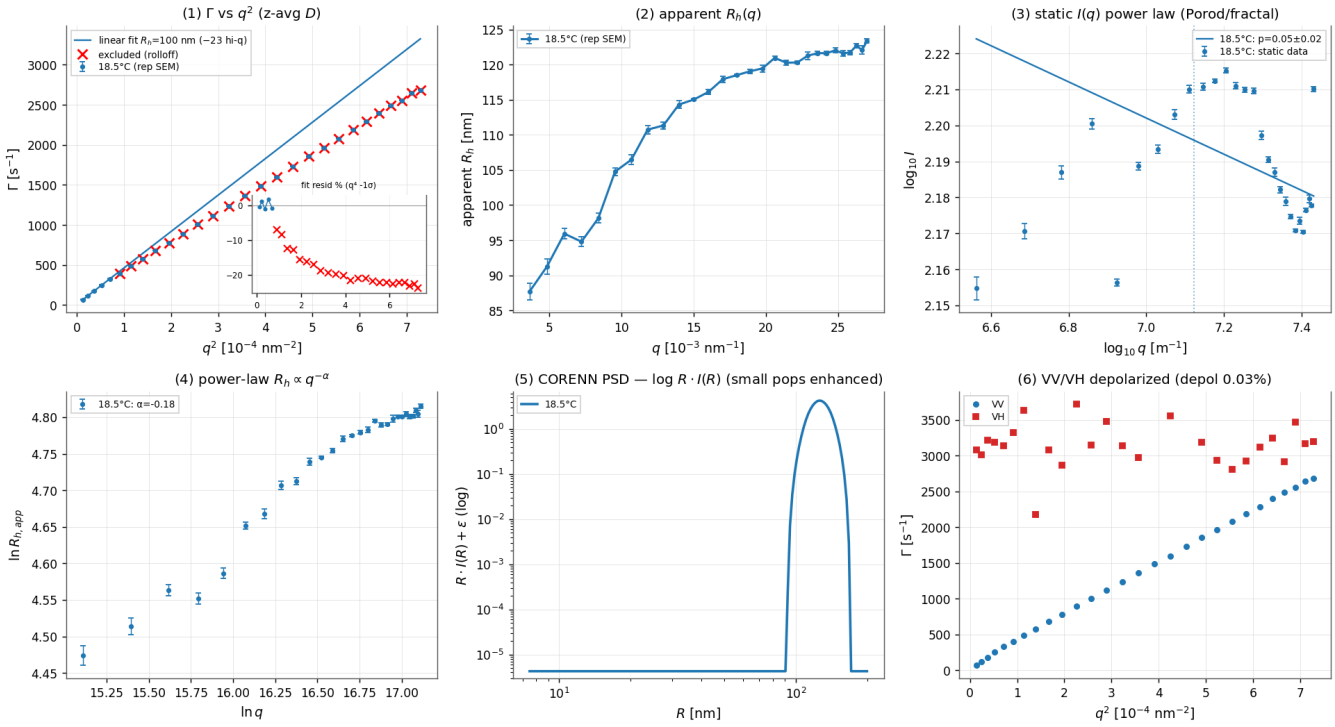
T = 11.5 °C


SISV_1% — all analysis panels at 11.5 °C (R_h,z=139 nm, PDI=0.25)
(1) Γ vs q^2 (z-avg D)
linear fit R_h=137 nm
11.5°C (rep SEM)
fit resid % (q^4 -31σ)
q^2 [10^{-4} nm^{-2}]
Γ [s^{-1}]
(2) apparent $R_h(q)$
11.5°C (rep SEM)
apparent R_h [nm]
q [10^{-3} nm^{-1}]
(3) static $I(q)$ power law (Porod/fractal)
11.5°C: p=-0.05±0.01
11.5°C: static data
log$_{10}$ I
log$_{10}$ q [m^{-1}]
(4) power-law $R_h \propto q^{-\alpha}$
11.5°C: α=-0.13
ln $R_{h,app}$
ln q
(5) CORENN PSD — log $R \cdot I(R)$ (small pops enhanced)
11.5°C
11.5°C peak 68%
$R \cdot I(R) + \varepsilon$ (log)
R [nm]
(6) VV/VH depolarized (depol 0.03%)
VV
VH
Γ [s^{-1}]
q^2 [10^{-4} nm^{-2}]

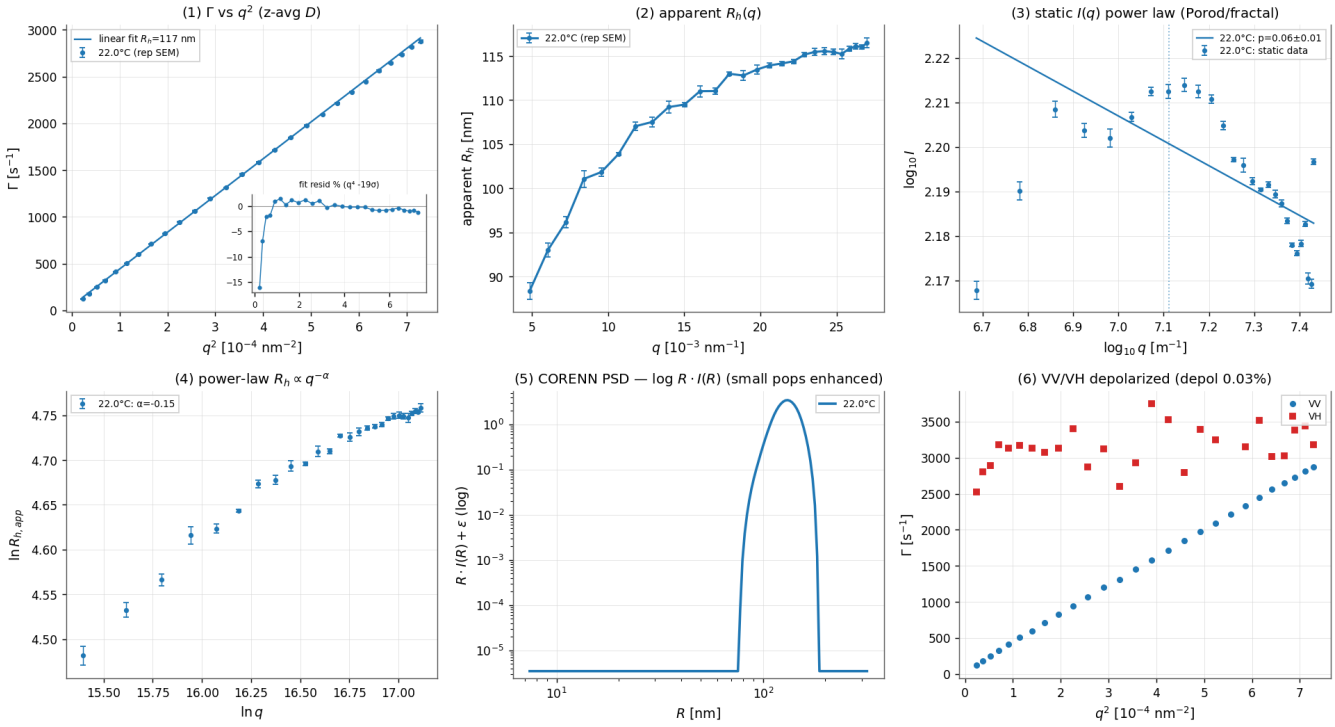
T = 15.0 °C


SISV_1% — all analysis panels at 15.0 °C (R_h,z=132 nm, PDI=0.22)
(1) Γ vs q^2 (z-avg D)
linear fit R_h=132 nm
15.0°C (rep SEM)
fit resid % (q^4 -3σ)
q^2 [10^{-4} nm^{-2}]
Γ [s^{-1}]
(2) apparent $R_h(q)$
15.0°C (rep SEM)
apparent R_h [nm]
q [10^{-3} nm^{-1}]
(3) static $I(q)$ power law (Porod/fractal)
15.0°C: p=-0.04±0.01
15.0°C: static data
log$_{10}$ I
log$_{10}$ q [m^{-1}]
(4) power-law $R_h \propto q^{-\alpha}$
15.0°C: α=-0.11
ln $R_{h,app}$
ln q
(5) CORENN PSD — log $R \cdot I(R)$ (small pops enhanced)
15.0°C
$R \cdot I(R) + \varepsilon$ (log)
R [nm]
(6) VV/VH depolarized (depol 0.03%)
VV
VH
Γ [s^{-1}]
q^2 [10^{-4} nm^{-2}]

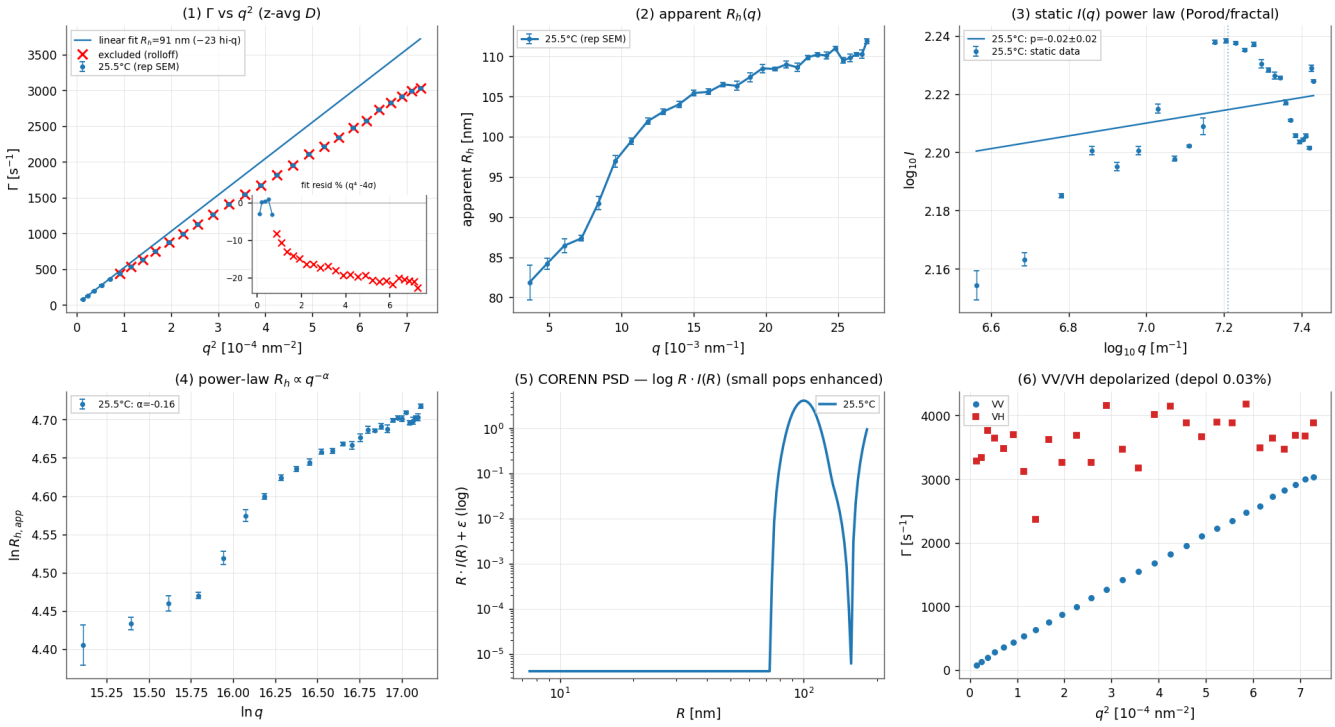
T = 18.5 °C


SISV_1% — all analysis panels at 18.5 °C (R_h,z=125 nm, PDI=0.28)
(1) Γ vs q^2 (z-avg D)
linear fit R_h=100 nm (−23 hi-q)
excluded (rolloff)
18.5°C (rep SEM)
fit resid % (q⁴ -1σ)
Γ [s⁻¹]
q^2 [10⁻⁴ nm⁻²]
(2) apparent $R_h(q)$
18.5°C (rep SEM)
apparent R_h [nm]
q [10⁻³ nm⁻¹]
(3) static $I(q)$ power law (Porod/fractal)
18.5°C: p=0.05±0.02
18.5°C: static data
log₁₀ I
log₁₀ q [m⁻¹]
(4) power-law $R_h \propto q^{-\alpha}$
18.5°C: α=-0.18
ln $R_{h,app}$
ln q
(5) CORENN PSD — log $R \cdot I(R)$ (small pops enhanced)
18.5°C
$R \cdot I(R) + \varepsilon$ (log)
R [nm]
(6) VV/VH depolarized (depol 0.03%)
VV
VH
Γ [s⁻¹]
q^2 [10⁻⁴ nm⁻²]

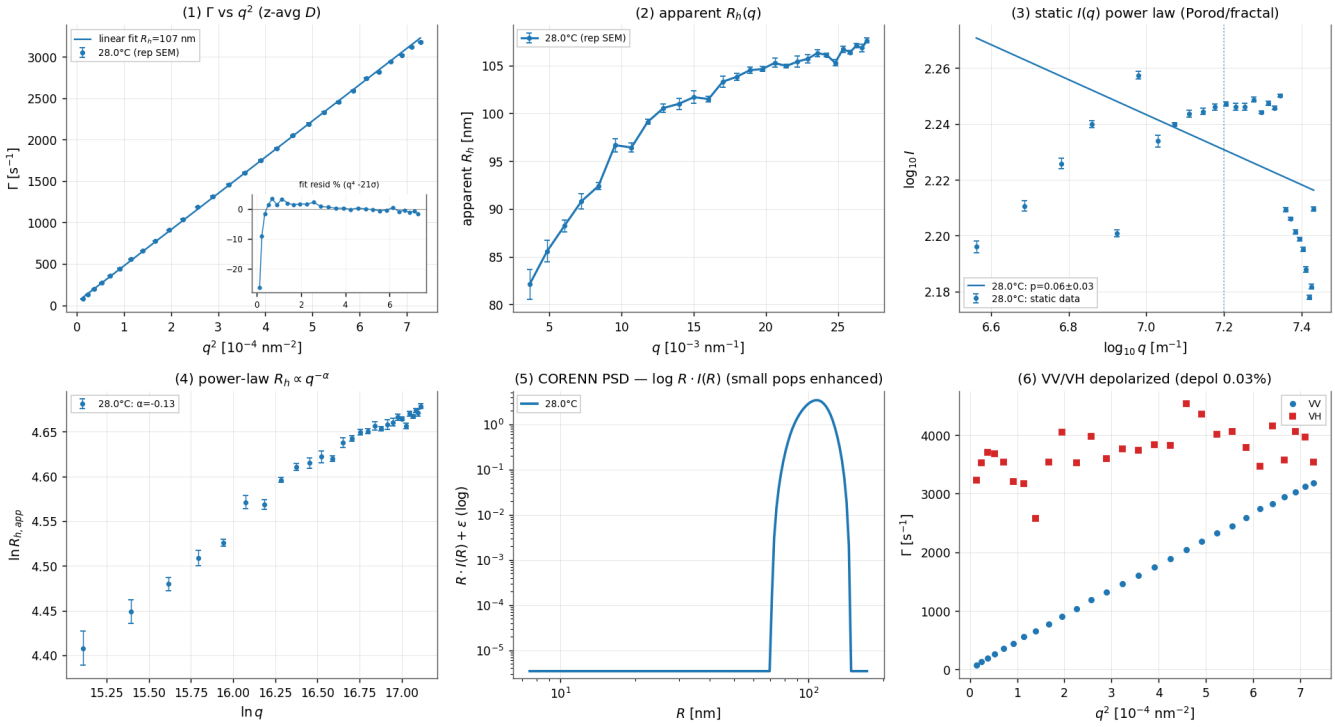
T = 22.0 °C


SISV_1% — all analysis panels at 22.0 °C (R_h,z=118 nm, PDI=0.25)
(1) Γ vs q^2 (z-avg D)
linear fit R_h=117 nm
22.0°C (rep SEM)
fit resid % (q⁴ -19σ)
Γ [s⁻¹]
q^2 [10⁻⁴ nm⁻²]
(2) apparent $R_h(q)$
22.0°C (rep SEM)
apparent R_h [nm]
q [10⁻³ nm⁻¹]
(3) static $I(q)$ power law (Porod/fractal)
22.0°C: p=0.06±0.01
22.0°C: static data
log₁₀ I
log₁₀ q [m⁻¹]
(4) power-law $R_h \propto q^{-\alpha}$
22.0°C: α=-0.15
ln $R_{h,app}$
ln q
(5) CORENN PSD — log $R \cdot I(R)$ (small pops enhanced)
22.0°C
$R \cdot I(R) + \varepsilon$ (log)
R [nm]
(6) VV/VH depolarized (depol 0.03%)
VV
VH
Γ [s⁻¹]
q^2 [10⁻⁴ nm⁻²]

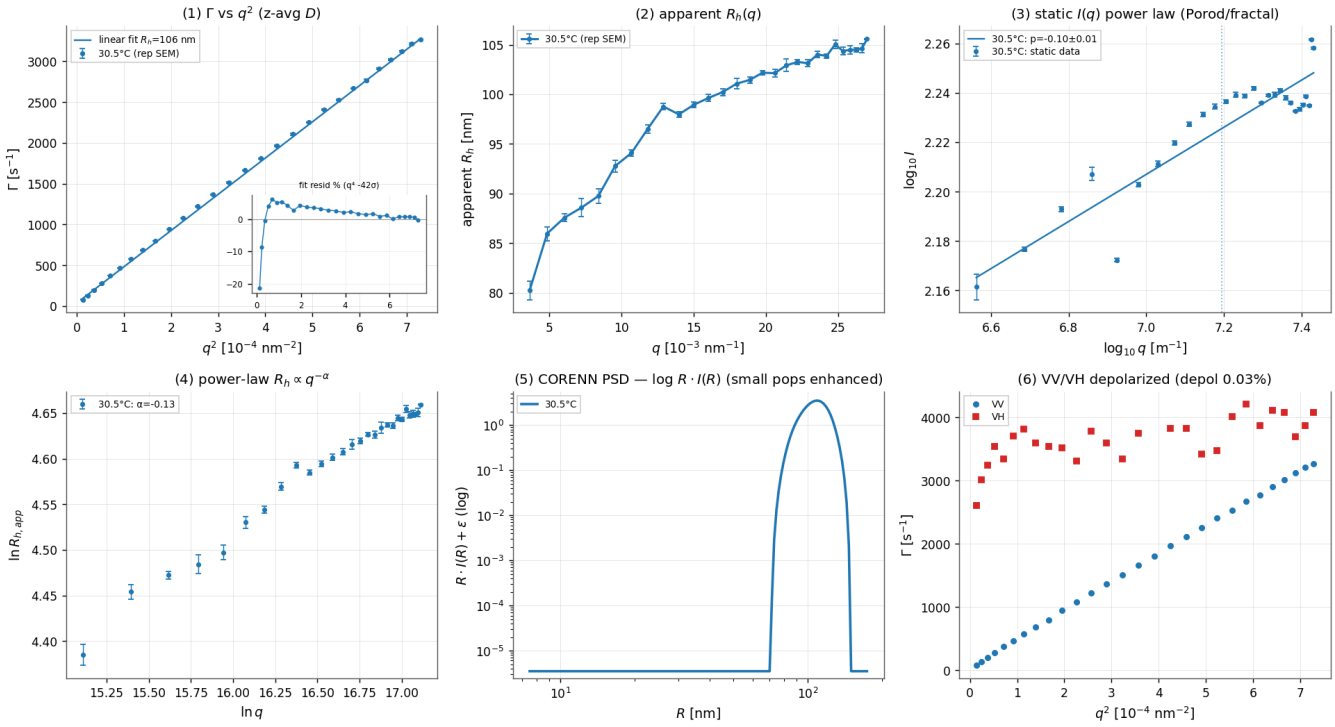
T = 25.5 °C


SISV_1% — all analysis panels at 25.5 °C (R_h,z=113 nm, PDI=0.28)
(1) Γ vs q^2 (z-avg D)
linear fit R_h=91 nm (−23 hi-q)
excluded (rolloff)
25.5°C (rep SEM)
fit resid % (q⁴ -4σ)
Γ [s⁻¹]
q^2 [10⁻⁴ nm⁻²]
(2) apparent $R_h(q)$
25.5°C (rep SEM)
apparent R_h [nm]
q [10⁻³ nm⁻¹]
(3) static $I(q)$ power law (Porod/fractal)
25.5°C: p=-0.02±0.02
25.5°C: static data
log₁₀ I
log₁₀ q [m⁻¹]
(4) power-law $R_h \propto q^{-\alpha}$
25.5°C: α=-0.16
ln $R_{h,app}$
ln q
(5) CORENN PSD — log $R \cdot I(R)$ (small pops enhanced)
25.5°C
$R \cdot I(R) + \varepsilon$ (log)
R [nm]
(6) VV/VH depolarized (depol 0.03%)
VV
VH
Γ [s⁻¹]
q^2 [10⁻⁴ nm⁻²]

T = 28.0 °C


SISV_1% — all analysis panels at 28.0 °C (R_h,z=108 nm, PDI=0.26)
(1) Γ vs q^2 (z-avg D)
linear fit R_h=107 nm
28.0°C (rep SEM)
fit resid % (q⁴ -21σ)
Γ [s⁻¹]
q^2 [10⁻⁴ nm⁻²]
(2) apparent $R_h(q)$
28.0°C (rep SEM)
apparent R_h [nm]
q [10⁻³ nm⁻¹]
(3) static $I(q)$ power law (Porod/fractal)
28.0°C: p=0.06±0.03
28.0°C: static data
log₁₀ I
log₁₀ q [m⁻¹]
(4) power-law $R_h \propto q^{-\alpha}$
28.0°C: α=-0.13
ln $R_{h,app}$
ln q
(5) CORENN PSD — log $R \cdot I(R)$ (small pops enhanced)
28.0°C
$R \cdot I(R) + \varepsilon$ (log)
R [nm]
(6) VV/VH depolarized (depol 0.03%)
VV
VH
Γ [s⁻¹]
q^2 [10⁻⁴ nm⁻²]

T = 30.5 °C

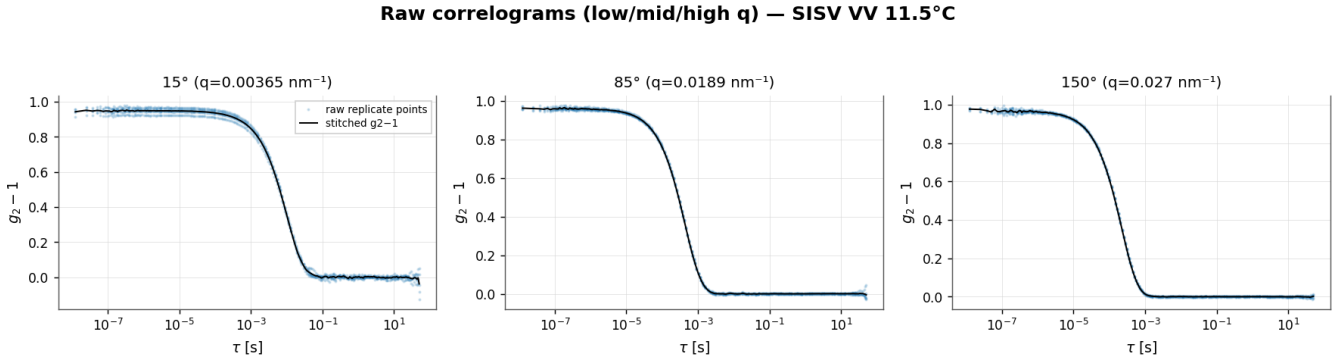


Data Inventory

One row per analysed condition (per-angle detail is in Per-Curve Fit Results).

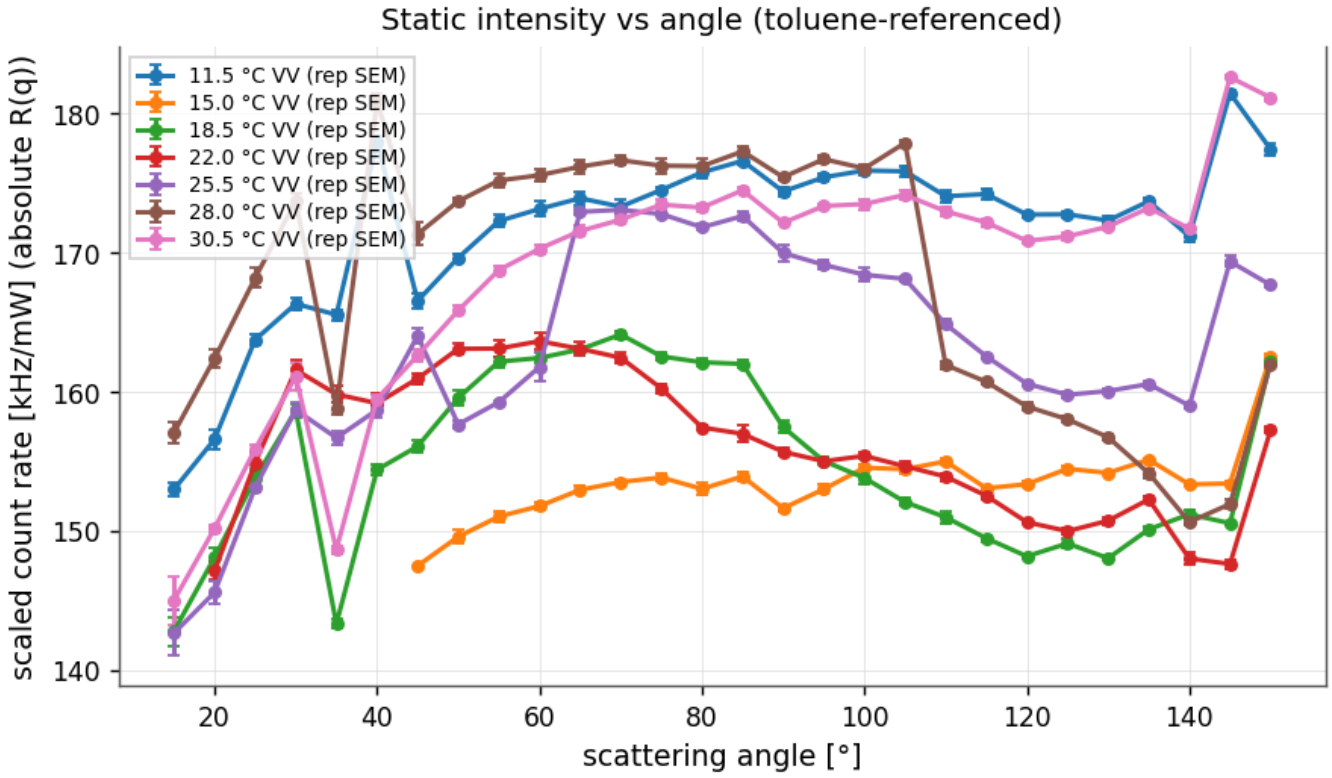
condition	channel	n angles (kept/total)	angle range	q range [nm^{-1}]	T [K]	scaled count rate [kHz/mW]	reps	missing meta
11.5 °C	VV	28/28	15–150°	0.00365–0.027	284.6	153–181	28	0
15.0 °C	VV	22/22	45–150°	0.0107–0.027	288.2	148–163	21	0
18.5 °C	VV	28/28	15–150°	0.00365–0.027	291.7	143–164	28	0
22.0 °C	VV	27/27	20–150°	0.00485–0.027	295.1	147–164	27	0
25.5 °C	VV	28/28	15–150°	0.00365–0.027	298.6	143–173	28	0
28.0 °C	VV	28/28	15–150°	0.00365–0.027	301.2	151–181	28	0
30.5 °C	VV	28/28	15–150°	0.00365–0.027	303.6	145–183	28	0

Raw Correlograms



Raw VV correlograms at low/mid/high q (11.5 °C): the stitched multi-tau correlation (black) over the semi-transparent raw replicate points.

Count Rates And Static Intensities (absolute, toluene-referenced)



Scaled count rate (absolute Rayleigh ratio) vs angle, all temperatures (raw data).

Per-Curve Fit Results

Per-curve fits — 11.5 °C (VV)

Per-angle 2nd-order cumulant on the leading decay (g1_min=0.6); $R_{\mathrm{h,app}} = \mathrm{kT}\ \mathrm{q}^2/(6\pi\eta\ \Gamma)$.

angle [°]	q [nm^{-1}]	Γ [1/s]	μ_2	PDI	β	$R_{\mathrm{h,app}}$ [nm]	reps	accepted
15	0.00365	54.5	1.3e+03	0.42	0.949	108	5	yes
20	0.00485	94.0	3.9e+03	0.44	0.979	111	5	yes
25	0.00605	141.3	7.7e+03	0.39	0.942	115	5	yes
30	0.00723	200.9	1.5e+04	0.37	0.986	116	5	yes
35	0.0084	265.2	2.0e+04	0.28	0.977	118	5	yes
40	0.00956	333.2	3.3e+04	0.29	0.998	122	5	yes
45	0.0107	410.4	5.0e+04	0.29	0.952	124	5	yes
50	0.0118	492.8	6.7e+04	0.27	0.937	126	5	yes
55	0.0129	581.6	8.0e+04	0.24	0.942	127	5	yes
60	0.014	671.9	1.0e+05	0.22	0.944	129	5	yes
65	0.015	768.6	1.3e+05	0.22	0.952	130	5	yes
70	0.016	881.2	1.8e+05	0.23	0.960	129	5	yes
75	0.017	976.1	2.2e+05	0.23	0.970	132	5	yes
80	0.018	1074.5	2.6e+05	0.23	0.959	133	5	yes
85	0.0189	1183.1	2.9e+05	0.21	0.956	134	5	yes
90	0.0198	1290.8	3.1e+05	0.19	0.958	134	5	yes
95	0.0206	1399.7	4.0e+05	0.20	0.956	135	5	yes
100	0.0214	1510.5	5.0e+05	0.22	0.959	135	5	yes
105	0.0222	1607.3	5.5e+05	0.21	0.964	136	5	yes
110	0.0229	1707.2	5.9e+05	0.20	0.960	136	5	yes
115	0.0236	1798.8	6.9e+05	0.21	0.959	137	5	yes
120	0.0242	1905.9	7.1e+05	0.20	0.947	136	5	yes
125	0.0248	1986.9	8.5e+05	0.21	0.954	137	5	yes
130	0.0253	2077.4	9.0e+05	0.21	0.950	137	5	yes
135	0.0258	2169.4	1.1e+06	0.23	0.942	136	5	yes
140	0.0263	2237.9	1.1e+06	0.22	0.944	137	5	yes
145	0.0267	2293.8	1.1e+06	0.21	0.976	137	5	yes
150	0.027	2346.1	1.1e+06	0.21	0.971	138	5	yes

Rejected Data And Failure Diagnostics

No angles or curves were rejected.

Assumption Register

assumption	value	source	affects	sensitivity checked?	comment
wavelength	632.8 nm	instrument default	q, D, R_h	no	—
refractive index	1.4	THF/dioxane 7:3, assumed	q	no	replace with measured
viscosity	0.47, 0.47, 0.47, 0.47, 0.47, 0.47, 0.47 mPa·s	recorded export (fixed 0.47 mPa·s)	R_h	yes	fixed at all temperatures — placeholder; THF thins on heating
q source	recalculated from angle	pipeline	all q-dependent fits	no	vendor 'Scattering vector [1/nm]' column is mislabelled (values are 1/m); q recomputed from angle, reported in nm^{-1}
static scale	absolute (toluene-referenced) — assumed, unverified	LSII toluene reference	SLS, R_g	no	toluene reference not confirmable from parsed export
form factor	homogeneous sphere (RGD)	pipeline	PSD radii, ρ check	yes	flag if $\rho \neq 0.78$ or depolarization present
cumulant g1_min	0.6	pipeline config	Γ, PDI	yes	leading-decay window
PSD radius grid	8–217 nm	dynamic observability + static q guardrail	PSD peaks	yes	upper=min(dynamic 217 nm, static 861 nm, default 5000 nm); limiting=dynamic; tau_max=0.0082 s, min field decay=0.223
PSD basis	24 B-spline control points	pipeline config	PSD smoothness	yes	—
regularization	GCV-selected λ=1.1e+02	pipeline	PSD width	yes	interior minimum required
baseline	free per fit (CORENN); leading-decay cumulant	pipeline config	Γ, PSD	yes	—

Quality Gates

Provisional report: **11 PASS, 2 WAIVE, 0 FAIL** of 13.

gate	status	note
q physically plausible / reproducible from angle	**PASS**	q recomputed from angle (vendor q column mislabeled 1/nm)
viscosity not a placeholder	**WAIVE**	same η recorded across temperatures — flagged placeholder
fit bounds stated for every fit	**PASS**	cumulant g1_min=0.6; CORENN tau_max=0.0082s, Rmax=217 nm (dynamic)
rejected curves/angles listed	**PASS**	0 rejected, tabulated
residual-structure test reported (not r^2 alone)	**PASS**	runs test + lag-1 autocorrelation
headline R_h carries a bootstrap CI	**PASS**	replicate + angle/model bootstrap
hold-out cross-validation reported	**PASS**	leave-one-angle-out
CORENN converged + well-posed regularization	**PASS**	converged=True, GCV interior=True, κ=5e+11
static scale confirmed absolute	**WAIVE**	toluene reference assumed (LSII default), not confirmable from parsed export
PSD basis (intensity/volume/number) stated	**PASS**	intensity-weighted primary, conversions noted
$\rho = R_g/R_h$ and particle model stated	**PASS**	not determined; sphere form factor assumed
multi-temperature size/viscosity ambiguity addressed	**PASS**	same-sample joint-temperature solve reported
figure captions state raw vs derived	**PASS**	captions included

Metadata

field	value
Report title	SISV_1% polymer (THF/DOX 7:3) — provisional DLS analysis report
Data source	data/SISV_1%
Report generator	reports/sisv_report.py (provisional)
Analyst / owner	automated pipeline
Dataset class	multi-angle, multi-temperature, polarized (VH/VV/3D)
Channels present	VV
Temperatures	284.6 K, 288.2 K, 291.7 K, 295.1 K, 298.6 K, 301.2 K, 303.6 K
Angles	15–150° (189 per condition)
q range	0.00365–0.027 nm^{-1}
Replicates	188 angle-sets with ≥2 replicates
Solvent constants	η per condition (see table); n=1.4, λ=632.8 nm (THF/dioxane 7:3, assumed)
Static scale	absolute (toluene-referenced) — assumed, unverified
Particle model	homogeneous sphere (assumed; ρ not determined — low-q S(q))
Primary model	CORENN multi-angle PSD (VV) + same-sample joint-temperature solve
Report limitations	viscosity placeholder; MAPCS replicates not exposed (angle bootstrap); sphere model assumed

Reproducibility Appendix

item	value
Command	`python reports/sisv_report.py`
Environment	python 3.13.5, numpy 2.2.5
Git commit (working tree)	91a3b68 (dirty)
Random seeds	bootstraps seed=0 (fixed)
Generated files	SISV.md, SISV.pdf, figures/sisv.png, figures/sisv_raw_rep.png, figures/sisv_slowmode.png, figures/sisv_weightings.png, figures/sisv_countrate.png, figures/sisv_highangle.png, SISV.sidecar.json, figures/sisv_T11p5.png, figures/sisv_T15p0.png, figures/sisv_T18p5.png, figures/sisv_T22p0.png, figures/sisv_T25p5.png, figures/sisv_T28p0.png, figures/sisv_T30p5.png